Operations Research

Midterm preparation: exercises and solutions

Exercise 1. A basic production model

A workshop produces divisible quantities of two products, A and B. Every unit produced is sold. The data per unit are:

	A	B
Processing time (hours)	2	1
Testing time (hours)	1	2
Profit (MAD)	50	40

The workshop has 8 processing hours and 7 testing hours available per day. Define decision variables and formulate an LP that maximizes daily profit.

Solution to Exercise 1

LP formulation

Let x_A and x_B be the numbers of units of A and B produced each day. The quantities are divisible, so $x_A, x_B \geq 0$ are continuous variables. The daily profit is $50x_A + 40x_B$ MAD.

Each unit of A uses 2 processing hours and 1 testing hour; each unit of B uses 1 processing hour and 2 testing hours. Thus the daily processing and testing limits give, respectively, $2x_A + x_B \leq 8$ and $x_A + 2x_B \leq 7$. The complete LP is

$$\begin{aligned} \max \quad & 50x_A + 40x_B \\ \text{s.t.} \quad & 2x_A + x_B \leq 8, \\ & x_A + 2x_B \leq 7, \\ & x_A, x_B \geq 0. \end{aligned}$$

The coefficients in both resource constraints are hours per unit; the objective is measured in MAD per day.

Exercise 2. An equivalent equational LP

Consider

$$\begin{aligned} \max \quad & 4x - 2y \\ \text{s.t.} \quad & 2x - y \leq 10, \\ & x + y \geq -5, \\ & x \geq -2, \qquad y \leq 0. \end{aligned}$$

Write an equivalent equational LP.

Solution to Exercise 2

Equational form

The lower bound on x suggests the shift $u = x + 2 \geq 0$. Since $y \leq 0$, set $v = -y \geq 0$. Thus $x = u - 2$ and $y = -v$.

The first inequality becomes $2u + v \leq 14$. We add a nonnegative slack s_1. The second becomes $u - v \geq -3$; we subtract a nonnegative surplus s_2. Hence an equivalent equational LP is

$$\begin{aligned} \max \quad & 4u + 2v - 8 \\ \text{s.t.} \quad & 2u + v + s_1 = 14, \\ & u - v - s_2 = -3, \\ & u, v, s_1, s_2 \geq 0. \end{aligned}$$

Exercise 3. Enumerating basic solutions

Let

$$A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{pmatrix}, \qquad b = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \qquad c = \begin{pmatrix} 3 \\ 2 \\ 4 \\ 7 \end{pmatrix}.$$

Consider $\max\{c^\top x : Ax = b,\ x \geq 0\}$.

(a) Determine the rank of A and justify your answer.

(b) Examine every selection of two columns. For each, state whether it is invertible; when it is, compute the full basic solution and determine whether it is feasible. Present the results in a table.

(c) Determine the optimal value and give one optimal solution.

Solution to Exercise 3

(a) Rank

Columns 1 and 2 form the 2×2 identity matrix, so they are linearly independent. Since A has only two rows, $\operatorname{rank}(A) = 2$.

(b) The six column choices

A basis requires two columns. There are exactly $\binom{4}{2} = 6$ possible choices. For each independent pair, set the other two variables to zero and solve $Ax = b$. For example, choosing columns 1 and 3 gives $x_3 = 2$ from the second equation and then $x_1 = 2$ from the first.

Columns	Invertible?	Full basic solution (x_1, x_2, x_3, x_4)	Feasible?
$1, 2$	Yes	$(4, 2, 0, 0)$	Yes
$1, 3$	Yes	$(2, 0, 2, 0)$	Yes
$1, 4$	Yes	$(2, 0, 0, 1)$	Yes
$2, 3$	Yes	$(0, -2, 4, 0)$	No
$2, 4$	Yes	$(0, -2, 0, 2)$	No
$3, 4$	No	—	Not a basis

The last pair is dependent because $A_4 = 2A_3$. The two infeasible basic solutions have $x_2 = -2$.

(c) Optimal value and one optimizer

The three feasible basic solutions have respective objective values 16, 14 and 13. To confirm that no other feasible point improves on 16, multiply the first equation of $Ax = b$ by 3 and the second by 2. This gives

$$3x_1 + 2x_2 + 5x_3 + 10x_4 = 16, \quad \text{hence} \quad c^\mathsf{T} x = 16 - x_3 - 3x_4 \leq 16.$$

The feasible point $(4, 2, 0, 0)$ attains this bound. Therefore

$$\boxed{z^\star = 16, \qquad x^\star = (4, 2, 0, 0).}$$

Exercise 4. Graphical solution

Consider

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 10, \\ & 3x_1 + x_2 \leq 12, \\ & x_1 \leq 4, \\ & x_1, x_2 \geq 0. \end{aligned}$$

(a) Determine which of the three displayed resource inequalities are redundant.

(b) Draw the feasible region and give the coordinates of every vertex.

(c) Solve the LP graphically and state the optimal solution and value.

Solution to Exercise 4

(a) Redundancy

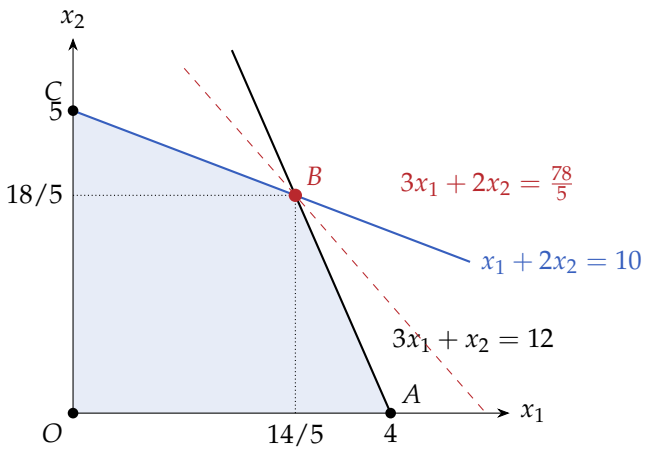
The inequality $x_1 \leq 4$ is redundant, since $3x_1 + x_2 \leq 12$ and $x_2 \geq 0$ imply $3x_1 \leq 12$. The first inequality is not redundant: $(0, 6)$ satisfies the other constraints but violates it. The second is not redundant: $(4, 3)$ satisfies the other constraints but violates it.

(b) Feasible region

Intersecting the boundary lines with the axes and with each other gives the four vertices

$$O = (0,0), \quad A = (4,0), \quad B = (\tfrac{14}{5}, \tfrac{18}{5}), \quad C = (0,5).$$

In particular, B is obtained by solving $x_1 + 2x_2 = 10$ and $3x_1 + x_2 = 12$. The shaded polygon below is the feasible region. Its vertices correspond to the BFSs after adding slacks.



(c) Graphical optimum

The objective levels are parallel lines $3x_1 + 2x_2 = k$. Moving these lines toward increasing k, the final contact with the polygon is at B. The vertex values are

$$z(O) = 0, \quad z(A) = 12, \quad z(B) = \tfrac{78}{5}, \quad z(C) = 10.$$

Therefore the unique optimizer is $x^\star = (14/5, 18/5)$ and $z^\star = 78/5$. The optimal level line meets the feasible polygon at this point only.

Exercise 5. A simplex calculation with three variables

Consider

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 + 3x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 \leq 10, \\ & x_1 + 3x_2 + 2x_3 \leq 15, \\ & x_1 + x_2 + 2x_3 \leq 12, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

(a) Solve the problem by simplex, starting from the slack-variable basis.

(b) State the optimal values of every decision and slack variable. Determine whether the optimum is unique and justify your answer.

Solution to Exercise 5

(a) Simplex solution

Introduce $s_1, s_2, s_3 \geq 0$ in constraint order. The initial tableau is

$$\begin{aligned} s_1 &= 10 - 2x_1 - x_2 - x_3 \\ s_2 &= 15 - x_1 - 3x_2 - 2x_3 \\ s_3 &= 12 - x_1 - x_2 - 2x_3 \\ z &= 5x_1 + 4x_2 + 3x_3 \end{aligned}$$

The initial BFS has all decision variables zero. We choose x_1 to enter because its reduced cost is positive. The bounds from the three rows are $x_1 \leq 5$, $x_1 \leq 15$ and $x_1 \leq 12$; therefore s_1 leaves. Solving the first row for x_1 and substituting gives

$$\begin{aligned} x_1 &= 5 - \frac{1}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2}s_1 \\ s_2 &= 10 - \frac{5}{2}x_2 - \frac{3}{2}x_3 + \frac{1}{2}s_1 \\ s_3 &= 7 - \frac{1}{2}x_2 - \frac{3}{2}x_3 + \frac{1}{2}s_1 \\ z &= 25 + \frac{3}{2}x_2 + \frac{1}{2}x_3 - \frac{5}{2}s_1 \end{aligned}$$

Now x_2 has the largest positive reduced cost. When $x_3 = s_1 = 0$, the three bounds on its increase are 10, 4 and 14. Thus s_2 leaves. The resulting tableau is

$$\begin{aligned} x_1 &= 3 - \frac{1}{5}x_3 - \frac{3}{5}s_1 + \frac{1}{5}s_2 \\ x_2 &= 4 - \frac{3}{5}x_3 + \frac{1}{5}s_1 - \frac{2}{5}s_2 \\ s_3 &= 5 - \frac{6}{5}x_3 + \frac{2}{5}s_1 + \frac{1}{5}s_2 \\ z &= 31 - \frac{2}{5}x_3 - \frac{11}{5}s_1 - \frac{3}{5}s_2 \end{aligned}$$

All nonbasic reduced costs are negative and all basic values are nonnegative. The tableau is optimal. This calculation uses *two pivots and three tableaux, including the initial one*.

(b) Full solution and uniqueness

Setting $x_3 = s_1 = s_2 = 0$ gives

$$\boxed{(x_1, x_2, x_3, s_1, s_2, s_3) = (3, 4, 0, 0, 0, 5), \qquad z^\star = 31.}$$

The final objective row shows that equality $z = 31$ is possible only if $x_3 = s_1 = s_2 = 0$. The remaining rows then determine every basic variable uniquely. Hence the optimum is unique.

Exercise 6. Solving through the dual

Consider

$$\begin{aligned} \max \quad & 6x_1 + 5x_2 + 4x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 \leq 14, \\ & x_1 + 2x_2 + 3x_3 \leq 16, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

(a) Write the dual LP.

(b) Solve the dual graphically.

(c) Determine a primal optimal solution and prove its optimality. Is it unique?

Solution to Exercise 6

(a) Dual formulation

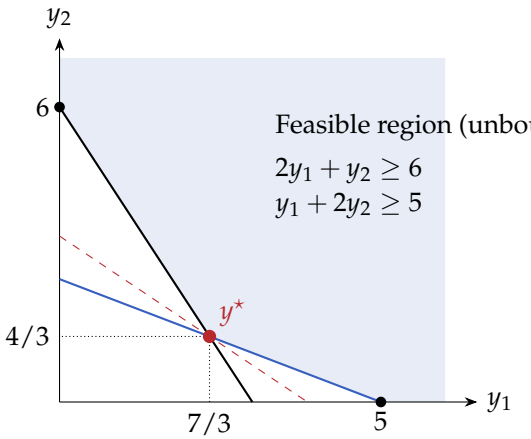
Let y_1, y_2 correspond to the first and second resource inequalities. The dual is

$$\begin{aligned} \min \quad & 14y_1 + 16y_2 \\ \text{s.t.} \quad & 2y_1 + y_2 \geq 6, \\ & y_1 + 2y_2 \geq 5, \\ & y_1 + 3y_2 \geq 4, \\ & y_1, y_2 \geq 0. \end{aligned}$$

(b) Graphical solution of the dual

The third inequality follows from $y_1 + 2y_2 \geq 5$ and $y_2 \geq 0$, so it does not change the feasible region. The region lies above the two remaining boundary lines. Its vertices are

$$(0, 6), \qquad (\tfrac{7}{3}, \tfrac{4}{3}), \qquad (5, 0).$$



The objective values at the vertices are 96, 54 and 70. Sliding $14y_1 + 16y_2 = k$ toward smaller k gives its last feasible contact at

$$\boxed{y^\star = (\tfrac{7}{3}, \tfrac{4}{3}), \qquad w^\star = 54.}$$

(c) A primal optimum and its uniqueness

At $y^\star$, the third dual constraint has positive slack:

$$y_1^\star + 3y_2^\star - 4 = \tfrac{7}{3} > 0.$$

Complementary slackness therefore requires $x_3 = 0$ at every primal optimum. Both dual variables are positive, so both primal resource constraints must be tight. We solve

$$2x_1 + x_2 = 14, \qquad x_1 + 2x_2 = 16,$$

which gives $x_1 = 4$ and $x_2 = 6$.

This candidate is nonnegative, uses exactly 14 and 16 resource units, and has value

$$6(4) + 5(6) + 4(0) = 54 = 14(\tfrac{7}{3}) + 16(\tfrac{4}{3}).$$

Primal and dual feasibility, together with equality of their objective values, prove optimality. The same complementary-slackness conditions are necessary for every primal optimum and have a unique solution. Thus

$$\boxed{x^\star = (4, 6, 0) \text{ is the unique primal optimum.}}$$

Exercise 7. A tableau, a basis and a direction

Consider

$$\begin{aligned} \max \quad & 4x_1 + 3x_2 + 2x_3 \\ \text{s.t.} \quad & x_1 + 2x_2 + x_3 \le 12, \\ & 3x_1 + x_2 + 2x_3 \le 16, \\ & x_1, x_2, x_3 \ge 0. \end{aligned}$$

Introduce slacks s_1, s_2 in constraint order. The following tableau is given:

$$\begin{array}{rcrrrr} x_1 & = & 4 & -\frac{3}{5}x_3 & +\frac{1}{5}s_1 & -\frac{2}{5}s_2 \\ x_2 & = & 4 & -\frac{1}{5}x_3 & -\frac{3}{5}s_1 & +\frac{1}{5}s_2 \\ \hline z & = & 28 & -x_3 & -s_1 & -s_2. \end{array}$$

(a) Give the associated full basic solution, in order $(x_1, x_2, x_3, s_1, s_2)$. Determine whether it is feasible and whether it is optimal.

(b) Give B and B^{-1} for the ordered basis (x_1, x_2).

(c) Find a dual optimal solution and verify it directly in the dual LP.

(d) Starting from the displayed basic solution, increase x_3 while keeping the other nonbasic variables zero. Determine the full direction vector, the largest feasible increase, and the point reached at that limit.

Solution to Exercise 7

(a) Reading the basic solution

The nonbasic variables are x_3, s_1, s_2. Setting them to zero gives

$$x = (4, 4, 0, 0, 0), \qquad z = 28.$$

All coordinates are nonnegative, so the basic solution is feasible. The objective row $z = 28 - x_3 - s_1 - s_2$ bounds every feasible objective by 28. Hence this BFS is optimal.

(b) Basis matrix and inverse

With the ordered basic variables (x_1, x_2), the corresponding columns of the equational matrix give

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, \qquad B^{-1} = \begin{pmatrix} -1/5 & 2/5 \\ 3/5 & -1/5 \end{pmatrix}.$$

Indeed $BB^{-1} = I_2$ and $B^{-1}(12, 16)^\mathsf{T} = (4, 4)^\mathsf{T}$.

(c) Dual certificate

The dual is

$$\min\{12y_1 + 16y_2 : y_1 + 3y_2 \geq 4,\ 2y_1 + y_2 \geq 3,\ y_1 + 2y_2 \geq 2,\ y \geq 0\}.$$

The basis gives $y^\mathsf{T} = (4, 3)B^{-1} = (1, 1)$. This vector satisfies the three inequalities with left-hand sides $4, 3, 3$, and its objective is $12 + 16 = 28$. It therefore certifies the primal optimum.

(d) Direction and endpoint

We use the basic-direction identity from the lecture. The column of the entering variable x_3 is $A_3 = (1, 2)^\mathsf{T}$. Set $d_3 = 1$ and $d_{s_1} = d_{s_2} = 0$. To preserve the equality $Ax = b$, the direction must satisfy

$$0 = Ad = Bd_B + A_3, \qquad \text{hence} \qquad d_B = -B^{-1}A_3.$$

Using the inverse computed in part (b), we obtain

$$d_B = -\begin{pmatrix} -1/5 & 2/5 \\ 3/5 & -1/5 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = -\begin{pmatrix} 3/5 \\ 1/5 \end{pmatrix}.$$

Therefore, in the specified full variable order,

$$d = (-\tfrac{3}{5}, -\tfrac{1}{5}, 1, 0, 0), \qquad x(\theta) = x + \theta d = (4 - \tfrac{3}{5}\theta,\ 4 - \tfrac{1}{5}\theta,\ \theta, 0, 0).$$

The two decreasing basic coordinates require $\theta \leq 4/(3/5) = 20/3$ and $\theta \leq 4/(1/5) = 20$. Thus the largest feasible increase is $\theta_{\max} = 20/3$, and the point reached is

$$\boxed{x(\theta_{\max}) = (0, \tfrac{8}{3}, \tfrac{20}{3}, 0, 0).}$$

Exercise 8. Funding divisible projects

A program can fund any fraction between 0 and 1 of each of five projects. Cost, staff time (in full-time equivalents), and benefit scale linearly with the funded fraction:

Project	A	B	C	D	E
Cost	4	6	5	3	2
Staff time	2	3	2	2	1
Benefit	8	12	11	6	5

The budget is 14 and at most 7 units of staff time are available. The funding fraction for C cannot exceed that for A. The funding fractions for B and D must sum to 1. The funding fraction for E cannot exceed the sum of those for B and C.

(a) Formulate a linear program maximizing total benefit.

(b) Consider the funding plan $(x_A, x_B, x_C, x_D, x_E) = (1, 0, 1, 1, 1)$. Check its feasibility and compute its benefit. As in the duality lecture, use a nonnegative combination of constraints to give a matching upper bound and certify optimality.

(c) Is this optimal plan unique? Justify your answer using the constraints that must be tight in part (b).

Solution to Exercise 8

(a) Linear programming formulation

For each project $J \in \{A, B, C, D, E\}$, let x_J denote its funded fraction. The LP is

$$\begin{aligned} \max \quad & 8x_A + 12x_B + 11x_C + 6x_D + 5x_E \\ \text{s.t.} \quad & 4x_A + 6x_B + 5x_C + 3x_D + 2x_E \le 14, \\ & 2x_A + 3x_B + 2x_C + 2x_D + x_E \le 7, \\ & x_C \le x_A, \\ & x_B + x_D = 1, \\ & x_E \le x_B + x_C, \\ & 0 \le x_J \le 1 \quad (J \in \{A, B, C, D, E\}). \end{aligned}$$

All variables are continuous. The coupling inequalities express the stated limits on funded fractions.

(b) Certifying the proposed funding plan

The proposed plan costs $4 + 5 + 3 + 2 = 14$ and uses $2 + 2 + 2 + 1 = 7$ staff-time units. It satisfies $x_C \le x_A$ ($1 \le 1$), $x_B + x_D = 1$ ($0 + 1 = 1$), $x_E \le x_B + x_C$ ($1 \le 0 + 1$), and all individual bounds. Its benefit is $8 + 11 + 6 + 5 = 30$.

To construct a dual certificate, compare the objective coefficients with twice the cost coefficients:

	A	B	C	D	E
benefit coefficient	8	12	11	6	5
$2 \times$ cost coefficient	8	12	10	6	4

The difference is 1 for C and 1 for E. Take the budget inequality with multiplier 2 and add $x_C \le 1$ and $x_E \le 1$, each with multiplier 1. All multipliers are nonnegative; the remaining constraints have multiplier zero. Their left-hand sides add to exactly the objective, so weak duality gives

$$8x_A + 12x_B + 11x_C + 6x_D + 5x_E \le 2(14) + 1 + 1 = 30 \quad \text{for every feasible } x.$$

The proposed plan attains 30; hence it is optimal, with $\boxed{z^\star = 30}$.

(c) Uniqueness

Equality in the bound from (b) requires the budget constraint to be tight and $x_C = x_E = 1$. The link $x_C \le x_A$ and bound $x_A \le 1$ then give $x_A = 1$. The equality $x_B + x_D = 1$ and the budget equality reduce to

$$11 + 6x_B + 3x_D = 14, \qquad x_B + x_D = 1.$$

They force $x_B = 0$ and $x_D = 1$. Thus every optimal solution is the one found in (b); the optimal plan is unique.

Exercise 9. A redundant inequality and several bases

For $\beta \in \mathbb{R}$, consider

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 10, \\ & x_1 + 3x_2 \leq 15, \\ & 5x_1 + 5x_2 \leq \beta, \\ & x_1, x_2 \geq 0. \end{aligned}$$

(a) Determine separately the values of β for which the LP is feasible and those for which the third inequality is redundant. Prove both answers.

(b) For $\beta = 30$, solve the LP by simplex.

Solution to Exercise 9

(a) Feasibility and redundancy

Since $x_1, x_2 \geq 0$, the left-hand side of the third inequality is nonnegative. If $\beta < 0$, the LP is infeasible. If $\beta \geq 0$, the origin is feasible. Thus feasibility holds exactly for $\beta \geq 0$.

Adding twice the first inequality to the second gives

$$5x_1 + 5x_2 \leq 35.$$

The point $(3, 4)$ satisfies the first two inequalities with equality and attains 35. Therefore the third inequality is redundant exactly when $\boxed{\beta \geq 35}$. For $\beta < 35$, this point is excluded, so the inequality is not redundant.

(b) Simplex for $\beta = 30$

The slack-variable tableau is

$$\begin{aligned} s_1 &= 10 - 2x_1 - x_2 \\ s_2 &= 15 - x_1 - 3x_2 \\ s_3 &= 30 - 5x_1 - 5x_2 \\ z &= 5x_1 + 4x_2 \end{aligned}$$

Choose x_1 to enter. Its bounds are $5, 15, 6$, so s_1 leaves. We obtain

$$\begin{aligned} x_1 &= 5 - \frac{1}{2}x_2 - \frac{1}{2}s_1 \\ s_2 &= 10 - \frac{5}{2}x_2 + \frac{1}{2}s_1 \\ s_3 &= 5 - \frac{5}{2}x_2 + \frac{5}{2}s_1 \\ z &= 25 + \frac{3}{2}x_2 - \frac{5}{2}s_1 \end{aligned}$$

Choose x_2 to enter. Its bounds are $10, 4, 2$, so s_3 leaves. The final tableau is

$$\begin{aligned} x_1 &= 4 - s_1 + \frac{1}{5}s_3 \\ s_2 &= 5 - 2s_1 + s_3 \\ x_2 &= 2 + s_1 - \frac{2}{5}s_3 \\ z &= 28 - s_1 - \frac{3}{5}s_3 \end{aligned}$$

The basic values are nonnegative and both nonbasic reduced costs are negative. Consequently

$$\boxed{(x_1, x_2, s_1, s_2, s_3) = (4, 2, 0, 5, 0), \qquad z^\star = 28.}$$

Exercise 10. Introducing a new activity

Consider

$$\begin{aligned} \max \quad & 6x_1 + 5x_2 + 3x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 \le 18, \\ & x_1 + 2x_2 + x_3 \le 18, \\ & x_1 + x_2 + 2x_3 \le 16, \\ & x_1, x_2, x_3 \ge 0. \end{aligned}$$

With slacks s_1, s_2, s_3 in constraint order, an optimal tableau is

$$\begin{array}{rcrrrr} x_1 &=& 6 & -\frac{1}{3}x_3 & -\frac{2}{3}s_1 & +\frac{1}{3}s_2 \\ x_2 &=& 6 & -\frac{1}{3}x_3 & +\frac{1}{3}s_1 & -\frac{2}{3}s_2 \\ s_3 &=& 4 & -\frac{4}{3}x_3 & +\frac{1}{3}s_1 & +\frac{1}{3}s_2 \\ \hline z &=& 66 & -\frac{2}{3}x_3 & -\frac{7}{3}s_1 & -\frac{4}{3}s_2. \end{array}$$

A new nonnegative activity u consumes 1, 2 and 2 units of the three resources, respectively, and contributes p per unit to the objective. No other data change.

(a) Write the modified LP and the tableau associated with the old ordered basis (x_1, x_2, s_3).

(b) Determine all values of p for which this basis remains optimal.

(c) Set $p = 10$. Continue simplex from that basis and find an optimal full solution and its value.

(d) Find a dual solution that certifies your answer for $p = 10$.

Solution to Exercise 10

(a) Adding the activity

The modified LP is

$$\begin{aligned} \max \quad & 6x_1 + 5x_2 + 3x_3 + pu \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 + u \le 18, \\ & x_1 + 2x_2 + x_3 + 2u \le 18, \\ & x_1 + x_2 + 2x_3 + 2u \le 16, \\ & x_1, x_2, x_3, u \ge 0. \end{aligned}$$

For the ordered basis (x_1, x_2, s_3),

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \qquad B^{-1} = \begin{pmatrix} 2/3 & -1/3 & 0 \\ -1/3 & 2/3 & 0 \\ -1/3 & -1/3 & 1 \end{pmatrix}.$$

The transformed new column is

$$B^{-1}(1,2,2)^{\mathsf{T}} = (0,1,1)^{\mathsf{T}}.$$

Hence increasing u leaves x_1 unchanged and decreases x_2, s_3 at rate 1. Its reduced cost is $p - (6,5,0)(0,1,1)^{\mathsf{T}} = p - 5$. The augmented tableau is

$$\begin{aligned} x_1 &= 6 - \tfrac{1}{3}x_3 - \tfrac{2}{3}s_1 + \tfrac{1}{3}s_2, \\ x_2 &= 6 - \tfrac{1}{3}x_3 - u + \tfrac{1}{3}s_1 - \tfrac{2}{3}s_2, \\ s_3 &= 4 - \tfrac{4}{3}x_3 - u + \tfrac{1}{3}s_1 + \tfrac{1}{3}s_2, \\ z &= 66 - \tfrac{2}{3}x_3 + (p-5)u - \tfrac{7}{3}s_1 - \tfrac{4}{3}s_2. \end{aligned}$$

(b) Range of optimality

The basic values remain $(6,6,4)$, so the old basis remains feasible. All previous reduced costs remain negative. The additional condition is $p - 5 \le 0$, giving $\boxed{p \le 5}$. No lower bound on p was imposed in the question. If $p > 5$, a small positive increase in u is feasible and improves the objective, so the old BFS is no longer optimal.

(c) Simplex for $p = 10$

The new variable has reduced cost 5 and enters. The bounds from x_2 and s_3 are $u \le 6$ and $u \le 4$, so s_3 leaves. Solving its row for u and substituting gives

$$\begin{aligned} x_1 &= 6 - \frac{1}{3}x_3 - \frac{2}{3}s_1 + \frac{1}{3}s_2 \\ x_2 &= 2 + x_3 - s_2 + s_3 \\ u &= 4 - \frac{4}{3}x_3 + \frac{1}{3}s_1 + \frac{1}{3}s_2 - s_3 \\ z &= 86 - \frac{22}{3}x_3 - \frac{2}{3}s_1 + \frac{1}{3}s_2 - 5s_3 \end{aligned}$$

Now s_2 has positive reduced cost $1/3$. As it increases, $x_2 = 2 - s_2$ is the only decreasing basic variable. Thus x_2 leaves after an increase of 2. We obtain

$$\begin{aligned} x_1 &= \frac{20}{3} - \frac{1}{3}x_2 - \frac{2}{3}s_1 + \frac{1}{3}s_3 \\ s_2 &= 2 - x_2 + x_3 + s_3 \\ u &= \frac{14}{3} - \frac{1}{3}x_2 - x_3 + \frac{1}{3}s_1 - \frac{2}{3}s_3 \\ z &= \frac{260}{3} - \frac{1}{3}x_2 - 7x_3 - \frac{2}{3}s_1 - \frac{14}{3}s_3 \end{aligned}$$

All nonbasic reduced costs are negative. In the order $(x_1, x_2, x_3, u, s_1, s_2, s_3)$, the optimal full solution and value are

$$\boxed{(\tfrac{20}{3}, 0, 0, \tfrac{14}{3}, 0, 2, 0), \qquad z^\star = \tfrac{260}{3}.}$$

(d) Dual certificate

The modified dual minimizes $18y_1 + 18y_2 + 16y_3$ over $y \ge 0$, subject to

$$\begin{aligned} 2y_1 + y_2 + y_3 &\ge 6, \quad & y_1 + 2y_2 + y_3 &\ge 5, \\ y_1 + y_2 + 2y_3 &\ge 3, \quad & y_1 + 2y_2 + 2y_3 &\ge 10. \end{aligned}$$

The final tableau in (c) has the ordered basis (x_1, s_2, u). Its columns and objective coefficients give

$$B = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & 2 \end{pmatrix}, \qquad c_B = \begin{pmatrix} 6 \\ 0 \\ 10 \end{pmatrix}.$$

The dual vector associated with a basis satisfies $y^\mathsf{T} = c_B^\mathsf{T} B^{-1}$, or equivalently $B^\mathsf{T} y = c_B$. Thus

$$2y_1 + y_2 + y_3 = 6, \qquad y_2 = 0, \qquad y_1 + 2y_2 + 2y_3 = 10.$$

Solving gives $\boxed{y = (2/3, 0, 14/3)}$. We can also read this from the final objective row: the coefficients of the nonbasic slacks s_1, s_3 are $-2/3, -14/3$, the negatives of y_1, y_3. The basic slack $s_2 = 2 > 0$ corresponds to $y_2 = 0$.

The four dual left-hand sides are $6, 16/3, 10, 10$, respectively, so y is dual feasible. Its value is

$$18(\tfrac{2}{3}) + 16(\tfrac{14}{3}) = \tfrac{260}{3},$$

which matches the feasible primal value and proves optimality.

Exercise 11. A resource change beyond the local interval

Consider

$$\begin{aligned} \max \quad & 4x_1 + 7x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \le 20, \\ & 3x_1 + x_2 \le 25, \\ & x_2 \le 9, \\ & x_1, x_2 \ge 0. \end{aligned}$$

With slacks in constraint order, an optimal tableau is

$$\begin{array}{rcrrr} x_1 & = & 6 & +\frac{1}{5}s_1 & -\frac{2}{5}s_2 \\ x_2 & = & 7 & -\frac{3}{5}s_1 & +\frac{1}{5}s_2 \\ s_3 & = & 2 & +\frac{3}{5}s_1 & -\frac{1}{5}s_2 \\ \hline z & = & 73 & -\frac{17}{5}s_1 & -\frac{1}{5}s_2. \end{array}$$

The first right-hand side changes from 20 to $20 + \delta$, where $\delta \in \mathbb{R}$. All other data remain unchanged.

(a) Determine the full interval of δ for which the ordered basis (x_1, x_2, s_3) remains optimal. Give the associated full solution and optimal value as functions of δ on this interval.

(b) For $\delta = 2$, state an optimal solution, its objective value and a dual certificate.

(c) For $\delta = 5$, determine whether the old basic solution is feasible. Find an optimal basis for the modified LP, give its tableau and state the new optimal solution and value.

Solution to Exercise 11

(a) Local interval and solution

With ordered basis (x_1, x_2, s_3), the first column of B^{-1} is $(-1/5, 3/5, -3/5)^\mathsf{T}$. Therefore replacing b by $b + \delta e_1$ changes the basic values to

$$x_1 = 6 - \tfrac{1}{5}\delta, \qquad x_2 = 7 + \tfrac{3}{5}\delta, \qquad s_3 = 2 - \tfrac{3}{5}\delta.$$

The nonbasic variables remain $s_1 = s_2 = 0$. The reduced costs do not depend on b and remain $-17/5, -1/5$. It remains to impose nonnegativity:

$$\delta \le 30, \qquad \delta \ge -\tfrac{35}{3}, \qquad \delta \le \tfrac{10}{3}.$$

Thus the complete interval for which this basis remains optimal is

$$\boxed{-\tfrac{35}{3} \le \delta \le \tfrac{10}{3}.}$$

On this interval the full solution is $(6 - \delta/5, 7 + 3\delta/5, 0, 0, 2 - 3\delta/5)$ and

$$\boxed{z^\star(\delta) = 73 + \tfrac{17}{5}\delta.}$$

(b) The change $\delta = 2$

Since 2 belongs to this interval, an optimal full solution is

$$(x_1, x_2, s_1, s_2, s_3) = (\tfrac{28}{5}, \tfrac{41}{5}, 0, 0, \tfrac{4}{5}), \qquad z^\star = \tfrac{399}{5}.$$

The dual vector $y = (17/5, 1/5, 0)$ is nonnegative and satisfies $y_1 + 3y_2 = 4$ and $2y_1 + y_2 + y_3 = 7$. Its value for the new right-hand side is

$$22(\tfrac{17}{5}) + 25(\tfrac{1}{5}) + 9(0) = \tfrac{399}{5}.$$

This certifies the answer.

(c) The change $\delta = 5$

The old basis would give $(x_1, x_2, s_3) = (5, 10, -1)$, so it is not feasible. To find a new candidate, impose the second and third resource inequalities as equalities:

$$3x_1 + x_2 = 25, \qquad x_2 = 9.$$

This gives $x_1 = 16/3$ and first slack $s_1 = 25 - 16/3 - 18 = 5/3$. All values are nonnegative. Consider the new ordered basis (x_1, x_2, s_1). Solving for these variables gives the tableau

$$\begin{aligned} x_1 &= \tfrac{16}{3} - \tfrac{1}{3}s_2 + \tfrac{1}{3}s_3, \\ x_2 &= 9 - s_3, \\ s_1 &= \tfrac{5}{3} + \tfrac{1}{3}s_2 + \tfrac{5}{3}s_3, \\ z &= \tfrac{253}{3} - \tfrac{4}{3}s_2 - \tfrac{17}{3}s_3. \end{aligned}$$

Both nonbasic reduced costs are negative, so this feasible tableau is optimal. Hence

$$\boxed{(x_1, x_2, s_1, s_2, s_3) = (\tfrac{16}{3}, 9, \tfrac{5}{3}, 0, 0), \qquad z^\star = \tfrac{253}{3}.}$$

As an independent certificate, $y = (0, 4/3, 17/3)$ is dual feasible and has value $25(4/3) + 9(17/3) = 253/3$.

Exercise 12. Changing a profit coefficient

Consider

$$\begin{aligned} \max \quad & 5x_1 + 3x_2 + 4x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \le 9, \\ & 2x_1 + x_2 + 3x_3 \le 15, \\ & x_1, x_2, x_3 \ge 0. \end{aligned}$$

An optimal tableau, with slacks s_1, s_2 in constraint order, is

$$\begin{array}{rcrrrr} x_1 & = & 6 & -2x_3 & +s_1 & -s_2 \\ x_2 & = & 3 & +x_3 & -2s_1 & +s_2 \\ \hline z & = & 39 & -3x_3 & -s_1 & -2s_2. \end{array}$$

Only the objective coefficient of x_1 changes, from 5 to $5 + \gamma$.

(a) Determine the full interval of γ for which the basis (x_1, x_2) remains optimal. Give the objective row of its tableau as a function of γ.

(b) At each endpoint of this interval, determine the complete set of optimal solutions.

(c) For $\gamma = 2$, solve the modified LP by continuing simplex from the displayed basis. Give a dual optimal solution as a certificate.

Solution to Exercise 12

(a) Range for the old basis

Changing the profit of x_1 adds γx_1 to the old objective. The constraint rows do not change, and setting the nonbasic variables $x_3 = s_1 = s_2 = 0$ still gives the feasible basic solution $(x_1, x_2) = (6, 3)$. Substitute the first tableau row, $x_1 = 6 - 2x_3 + s_1 - s_2$, into $z_\gamma = z_0 + \gamma x_1$:

$$\boxed{z_\gamma = 39 + 6\gamma + (-3 - 2\gamma)x_3 + (-1 + \gamma)s_1 + (-2 - \gamma)s_2.}$$

Since we maximize, this feasible basis is optimal when all three nonbasic coefficients in the objective row are nonpositive. They give

$$\begin{aligned} -3 - 2\gamma \le 0 &\iff \gamma \ge -\tfrac{3}{2}, \quad (x_3), \\ -1 + \gamma \le 0 &\iff \gamma \le 1, \quad (s_1), \\ -2 - \gamma \le 0 &\iff \gamma \ge -2, \quad (s_2). \end{aligned}$$

Their intersection is $\boxed{-\tfrac{3}{2} \le \gamma \le 1}$. Outside this interval, a positive reduced cost gives an improving move from $(6, 3)$: both basic coordinates are strictly positive, so a sufficiently small step is feasible.

(b) All optimal solutions at the endpoints

At $\gamma = -3/2$, the coefficient of x_3 is zero. The objective row reads

$$z = 30 - \tfrac{5}{2}s_1 - \tfrac{1}{2}s_2.$$

It bounds the objective by 30. Equality holds precisely when $s_1 = s_2 = 0$; x_3 may vary. Put $x_3 = t$ in the two constraint rows. Then $x_1 = 6 - 2t$ and $x_2 = 3 + t$, and nonnegativity requires $0 \le t \le 3$. Thus the complete set of full optimal solutions, in the order $(x_1, x_2, x_3, s_1, s_2)$, is

$$\boxed{\{(6 - 2t, 3 + t, t, 0, 0) : 0 \le t \le 3\}.}$$

At $\gamma = 1$, the coefficient of s_1 is zero and $z = 45 - 5x_3 - 3s_2$. Equality in $z \le 45$ requires $x_3 = s_2 = 0$. Put $s_1 = t$ in the constraint rows: $x_1 = 6 + t$ and $x_2 = 3 - 2t$. Nonnegativity gives $0 \le t \le 3/2$, so the complete set is

$$\boxed{\{(6 + t, 3 - 2t, 0, t, 0) : 0 \le t \le \tfrac{3}{2}\}.}$$

(c) Continuing simplex at $\gamma = 2$

For $\gamma = 2$ the old basis is still feasible, but its objective row is $z = 51 - 7x_3 + s_1 - 4s_2$. The positive coefficient of s_1 makes it an entering variable. Starting with $x_3 = s_2 = 0$ and increasing s_1, the tableau gives $x_1 = 6 + s_1$ and $x_2 = 3 - 2s_1$. Thus x_2 first reaches zero at $s_1 = 3/2$ and leaves the basis.

Solving the x_2 row for s_1 gives $s_1 = \frac{3}{2} - \frac{1}{2}x_2 + \frac{1}{2}x_3 + \frac{1}{2}s_2$. Substitute this into the other two rows to obtain

$$x_1 = \frac{15}{2} - \frac{1}{2}x_2 - \frac{3}{2}x_3 - \frac{1}{2}s_2$$
$$s_1 = \frac{3}{2} - \frac{1}{2}x_2 + \frac{1}{2}x_3 + \frac{1}{2}s_2$$
$$z = \frac{105}{2} - \frac{1}{2}x_2 - \frac{13}{2}x_3 - \frac{7}{2}s_2$$

With nonbasic variables $x_2 = x_3 = s_2 = 0$, the basic variables are $(x_1, s_1) = (15/2, 3/2)$, which are nonnegative. Every nonbasic coefficient in the new objective row is strictly negative, so this is the unique optimum:

$$\boxed{(x_1, x_2, x_3, s_1, s_2) = (\tfrac{15}{2}, 0, 0, \tfrac{3}{2}, 0), \qquad z^\star = \tfrac{105}{2}.}$$

For a dual certificate, the first slack is basic and positive, so $y_1 = 0$. The coefficient of the nonbasic slack s_2 in the final objective row is $-7/2$, giving $y_2 = 7/2$. The dual for $\gamma = 2$ is

$$\min\{9y_1 + 15y_2 : y_1 + 2y_2 \geq 7,\ y_1 + y_2 \geq 3,\ y_1 + 3y_2 \geq 4,\ y \geq 0\}.$$

The vector $y = (0, 7/2)$ satisfies its three inequalities and has value $15(7/2) = 105/2$. This equals the primal value and independently certifies optimality.

Exercise 13. A parameter changes the status of an LP

For $\alpha \in \mathbb{R}$, consider

$$\begin{aligned} \max \quad & (\alpha - 1)x_1 + x_2 \\ \text{s.t.} \quad & -x_1 + x_2 \leq -1, \\ & x_2 \leq \alpha, \\ & x_1, x_2 \geq 0. \end{aligned}$$

(a) Determine the values of α for which the primal LP has (i) no feasible solution, (ii) a finite maximum, or (iii) arbitrarily large objective values. Justify each case.

(b) For every α in case (ii), compute the maximum value and describe *all* pairs (x_1, x_2) that attain it.

(c) Write the dual LP. Determine the values of α for which the dual (i) has no feasible solution, (ii) is feasible but its objective can decrease without bound, or (iii) has a finite minimum. In case (iii), give a primal optimal solution and a dual optimal solution, and verify that their objective values agree.

(d) Determine the values of α for which the primal feasible region is unbounded although the maximum objective value is finite. For which of these values does the set of optimal solutions contain points with arbitrarily large x_1?

Solution to Exercise 13

(a) Classification of the primal

The constraints are equivalent to

$$0 \le x_2 \le \alpha, \qquad x_1 \ge x_2 + 1.$$

If $\alpha < 0$, the interval for x_2 is empty and the primal is infeasible. If $\alpha \ge 0$, the point $(1, 0)$ is feasible.

For $0 \le \alpha < 1$, the coefficient of x_1 is negative. At a fixed x_2, the objective is therefore maximized by the smallest permitted value $x_1 = x_2 + 1$. Substitution gives

$$z = (\alpha - 1)(x_2 + 1) + x_2 = \alpha x_2 + \alpha - 1.$$

This expression has a finite maximum over $0 \le x_2 \le \alpha$. For $\alpha = 1$, the objective is simply $x_2 \le 1$, so again the optimum is finite.

If $\alpha > 1$, the feasible sequence $(x_1, x_2) = (1 + t, 0)$, $t \ge 0$, has objective $(\alpha - 1)(1 + t) \to +\infty$. Hence

$\alpha < 0$	infeasible
$0 \le \alpha \le 1$	finite optimum
$\alpha > 1$	unbounded objective.

(b) Values and all optimal solutions

For $0 < \alpha < 1$, the last expression increases strictly with x_2, so $x_2 = \alpha$ and $x_1 = \alpha + 1$. At $\alpha = 0$, the only possible x_2 is 0 and maximizing $-x_1$ gives $x_1 = 1$. Thus for all $0 \le \alpha < 1$,

$$\operatorname{argmax} z = \{(\alpha + 1, \alpha)\}, \qquad z^\star = \alpha^2 + \alpha - 1.$$

At $\alpha = 1$, every point with $x_2 = 1$ and $x_1 \ge 2$ is optimal, with value 1. Therefore

$$\boxed{\operatorname{argmax} z = \{(x_1, 1) : x_1 \ge 2\} \quad \text{when } \alpha = 1.}$$

(c) The dual and its status

The dual is

$$\begin{aligned} \min \quad & -y_1 + \alpha y_2 \\ \text{s.t.} \quad & -y_1 \geq \alpha - 1, \\ & y_1 + y_2 \geq 1, \\ & y_1, y_2 \geq 0. \end{aligned}$$

If $\alpha > 1$, the first inequality contradicts $y_1 \geq 0$, so the dual is infeasible. If $\alpha < 0$, choose $y_1 = 0$ and $y_2 = 1 + t$, $t \geq 0$. This is dual feasible and has value $\alpha(1 + t) \to -\infty$, so the dual is unbounded below.

For $0 \leq \alpha \leq 1$, take

$$x = (\alpha + 1, \alpha), \qquad y = (1 - \alpha, \alpha).$$

Both points are feasible, and their common objective value is

$$(\alpha - 1)(\alpha + 1) + \alpha = -(1 - \alpha) + \alpha^2 = \alpha^2 + \alpha - 1.$$

They form an optimal primal–dual pair. Consequently the status table is

Parameter	Primal	Dual
$\alpha < 0$	Infeasible	Unbounded below
$0 \leq \alpha \leq 1$	Finite optimum	Finite optimum
$\alpha > 1$	Unbounded above	Infeasible

(d) Unbounded sets versus unbounded objectives

Whenever $\alpha \geq 0$, the ray $\{(1 + t, 0) : t \geq 0\}$ lies in the primal feasible set, so that set is unbounded. Its objective is bounded above exactly for $0 \leq \alpha \leq 1$. Among those values, the set of optimizers is unbounded only at $\alpha = 1$; for $0 \leq \alpha < 1$ the optimizer is unique.

Exercise 14. Why an optimal BFS exists

Let

$$P = \{x \in \mathbb{R}^n : Ax = b,\ x \geq 0\}, \qquad A \in \mathbb{R}^{m \times n}, \qquad \operatorname{rank}(A) = m.$$

Assume that P is nonempty and that $c^\mathsf{T} x$ is bounded above on P. Write A_j for column j of A.

Give proofs of the following statements. You may use elementary linear algebra and the definition of a BFS, but not the theorem asserting the existence of an optimal BFS.

(a) A point $x \in P$ is a basic feasible solution if and only if the indexed family of columns

$$(A_j)_{j : x_j > 0}$$

is linearly independent.

(b) For every $x \in P$, there exists a basic feasible solution $\widehat{x} \in P$ satisfying $c^\mathsf{T} \widehat{x} \geq c^\mathsf{T} x$.

(c) The maximum of $c^\mathsf{T} x$ over P is attained at a basic feasible solution.

Solution to Exercise 14

(a) Characterizing a BFS by its positive coordinates

For a feasible point x, let $S = \{j : x_j > 0\}$.

First suppose x is a BFS represented by a basis index set I. Every nonbasic coordinate is zero, so $S \subseteq I$. The basis columns are independent; consequently their subfamily indexed by S is independent as well.

Conversely, suppose $(A_j)_{j \in S}$ is independent. Since $\operatorname{rank}(A) = m$, this family can be extended, using other columns of A, to m independent columns indexed by a set I. All positive coordinates of x belong to I, so $x_j = 0$ for $j \notin I$. The remaining coordinates satisfy $A_I x_I = b$. Since A_I is invertible, $x_I = A_I^{-1} b$. Thus x is the basic solution associated with I; it is feasible by assumption, hence is a BFS.

(b) Reducing the support without decreasing the objective

Take $x \in P$. If its positive-support columns are independent, part (a) already gives the result. Otherwise there is a nonzero vector d supported on S such that $Ad = 0$.

We claim that its sign can be chosen so that

$$c^\mathsf{T} d \geq 0 \qquad \text{and} \qquad d_j < 0 \text{ for at least one } j.$$

If $c^\mathsf{T} d = 0$, choose the sign of any nonzero component to be negative. If $c^\mathsf{T} d \neq 0$, choose the sign so that $c^\mathsf{T} d > 0$. With this sign, d cannot be nonnegative: otherwise $x + td \in P$ for every $t \geq 0$ and $c^\mathsf{T}(x + td) \to +\infty$, contradicting boundedness above.

Now define

$$t_\star = \min_{j : d_j < 0} \frac{x_j}{-d_j} > 0, \qquad x' = x + t_\star d.$$

The minimum is over a nonempty finite set. Each of its numerators is positive, because d is supported on S. By construction $x' \geq 0$, and

$$Ax' = Ax + t_\star Ad = b, \qquad c^\mathsf{T} x' = c^\mathsf{T} x + t_\star c^\mathsf{T} d \geq c^\mathsf{T} x.$$

At least one previously positive coordinate becomes zero. No previously zero coordinate becomes positive, because $d_j = 0$ outside S. Thus the support strictly decreases.

Repeat this procedure whenever the current support columns are dependent. It must stop after finitely many steps because the support size decreases each time. The final point $\widehat{x}$ has independent support columns, hence is a BFS by part (a), and its objective has never decreased. Therefore $c^\mathsf{T} \widehat{x} \geq c^\mathsf{T} x$.

(c) Attainment at a BFS

Part (b), applied to any point of the nonempty set P, proves that at least one BFS exists. There are at most $\binom{n}{m}$ choices of basis indices, and each invertible choice determines at most one basic solution. Thus the set of BFSs is finite and nonempty.

Choose a BFS $x^\star$ with the largest objective value among them. For an arbitrary $x \in P$, part (b) supplies a BFS $\widehat{x}$ with

$$c^\mathsf{T} x \le c^\mathsf{T} \widehat{x} \le c^\mathsf{T} x^\star.$$

Since this holds for every feasible x, the point $x^\star$ attains the maximum over P.

A nonempty equational LP bounded above has an optimal BFS.

This argument proves attainment directly; it does not assume in advance that an optimal point exists. Boundedness above was used in part (b) to rule out a strictly improving nonnegative ray.

Exercise 15. Uniform approximation by a line

Three measurements are given:

t_i	0	1	2
y_i	1	4	2

We wish to approximate them by a line $a + bt$, where $a, b \in \mathbb{R}$. The quality of a line is measured by its largest absolute error:

$$E(a,b) = \max_{i \in \{1,2,3\}} |a + bt_i - y_i|.$$

(a) Formulate $\min_{a,b \in \mathbb{R}} E(a,b)$ as an LP with a linear objective and linear constraints. No maximum or absolute-value operator may remain in the formulation.

(b) Determine an optimal line and the minimum possible error.

(c) Write the dual of your LP and find a dual feasible solution certifying the value obtained in part (b).

(d) Is the optimal line unique? Prove your answer.

Solution to Exercise 15

(a) LP formulation

Introduce $\varepsilon \geq 0$ to bound every absolute error. An equivalent LP is

$$\begin{aligned} \min \quad & \varepsilon \\ \text{s.t.} \quad & a - 1 \leq \varepsilon, \qquad 1 - a \leq \varepsilon, \\ & a + b - 4 \leq \varepsilon, \qquad 4 - a - b \leq \varepsilon, \\ & a + 2b - 2 \leq \varepsilon, \qquad 2 - a - 2b \leq \varepsilon, \\ & a, b \in \mathbb{R}, \qquad \varepsilon \geq 0. \end{aligned}$$

For any fixed a, b, the smallest feasible ε is exactly $E(a, b)$, so this formulation preserves the minimum and the optimal lines.

(b) An optimal line

We first derive a lower bound on the LP objective by taking a nonnegative combination of three constraints from part (a):

$$\begin{aligned} a - 1 &\leq \varepsilon \quad \text{(first observation)}, \\ 2(4 - a - b) &\leq 2\varepsilon \quad \text{(second observation)}, \\ a + 2b - 2 &\leq \varepsilon \quad \text{(third observation)}. \end{aligned}$$

Adding them cancels both a and b and yields

$$5 \leq 4\varepsilon, \qquad \text{so every feasible line has } \varepsilon \geq \tfrac{5}{4}.$$

To find a line attaining this bound, set $\varepsilon = 5/4$. All three inequalities used above must then hold with equality: their nonnegative slacks sum to $4\varepsilon - 5 = 0$. The first two equalities give

$$a - 1 = \tfrac{5}{4} \implies a = \tfrac{9}{4}, \qquad 4 - a - b = \tfrac{5}{4} \implies b = \tfrac{1}{2}.$$

The third equality also holds: $a + 2b - 2 = 5/4$. The three signed residuals are therefore $(5/4, -5/4, 5/4)$, so all six inequalities in part (a) are satisfied. The lower bound is attained, and hence

$$\boxed{a = \tfrac{9}{4}, \qquad b = \tfrac{1}{2}, \qquad E_{\min} = \tfrac{5}{4}.}$$

The weights $(1, 2, 1)/4$ in the bound are also the dual multipliers used in part (c).

(c) Dual and certificate

To apply the dual-construction rules for a minimization LP, write the two constraints for observation i as

$$\varepsilon - a - bt_i \geq -y_i, \qquad \varepsilon + a + bt_i \geq y_i.$$

Associate nonnegative dual variables u_i and v_i with these constraints, respectively. Since a and b are free, their two dual constraints are equalities. Since $\varepsilon \geq 0$ has objective coefficient 1, its dual constraint is $\sum_i (u_i + v_i) \leq 1$. The dual is therefore

$$\begin{aligned} \max \quad & \sum_{i=1}^{3} y_i (v_i - u_i) \\ \text{s.t.} \quad & \sum_{i=1}^{3} (u_i - v_i) = 0, \\ & \sum_{i=1}^{3} t_i (u_i - v_i) = 0, \\ & \sum_{i=1}^{3} (u_i + v_i) \leq 1, \qquad u_i, v_i \geq 0. \end{aligned}$$

(c) Verifying the dual certificate

Take $u = (1/4, 0, 1/4)$ and $v = (0, 1/2, 0)$. Then

$$\sum_i (u_i - v_i) = \tfrac{1}{4} - \tfrac{1}{2} + \tfrac{1}{4} = 0,$$

$$\sum_i t_i(u_i - v_i) = 0 - \tfrac{1}{2} + 2(\tfrac{1}{4}) = 0, \qquad \sum_i (u_i + v_i) = 1.$$

The dual objective is

$$1(-\tfrac{1}{4}) + 4(\tfrac{1}{2}) + 2(-\tfrac{1}{4}) = \tfrac{5}{4}.$$

It matches the feasible primal value. Weak duality therefore certifies optimality.

(d) Uniqueness

Any optimal line has $\varepsilon = 5/4$ and must make the three inequalities combined in part (b) tight. The first gives $a = 9/4$; the second then gives $b = 1/2$. Consequently no other line can attain the minimum error, and the optimal line is unique.

Exercise 16. Reduced costs: what can and cannot be concluded

Consider the maximization LP

$$\max\{c^{\mathsf{T}}x : Ax = b,\ x \geq 0\}, \qquad A \in \mathbb{R}^{m\times n}, \quad \operatorname{rank}(A) = m.$$

Let B be an invertible basis matrix, let $\mathcal{N}$ be its set of nonbasic indices, and let x be its associated basic solution. Assume that x is feasible. Denote the reduced cost of a nonbasic variable x_j by $\bar{c}_j$.

(a) Prove that $\bar{c}_j \leq 0$ for every $j \in \mathcal{N}$ implies that x is optimal.

(b) Prove that $\bar{c}_j < 0$ for every $j \in \mathcal{N}$ implies that x is the unique optimal solution.

(c) Suppose x is nondegenerate. Prove that optimality of x implies $\bar{c}_j \leq 0$ for every $j \in \mathcal{N}$.

(d) Construct an explicit LP and a basis whose associated basic solution is feasible, degenerate and optimal, but for which at least one nonbasic reduced cost is positive. Verify every claimed property.

(e) Suppose x is nondegenerate and optimal, and some nonbasic reduced cost is zero. Prove that the LP has infinitely many optimal solutions.

Solution to Exercise 16

For any feasible vector w, partition its coordinates into basic and nonbasic components. The equality constraints imply

$$w_B = B^{-1}b - B^{-1}A_{\mathcal{N}}w_{\mathcal{N}}.$$

Substitution into the objective gives the identity

$$\boxed{c^{\mathsf{T}}w = c_B^{\mathsf{T}}B^{-1}b + \sum_{j\in\mathcal{N}} \bar{c}_j w_j, \qquad \bar{c}_j = c_j - c_B^{\mathsf{T}}B^{-1}A_j.}$$

The constant is the objective of the specified BFS x.

(a) Nonpositive reduced costs are sufficient

If all $\bar{c}_j \leq 0$, every feasible w satisfies $w_j \geq 0$, so the displayed identity gives $c^{\mathsf{T}}w \leq c^{\mathsf{T}}x$. Since x is feasible, it is optimal.

(b) Strictly negative reduced costs imply uniqueness

If all nonbasic reduced costs are strictly negative, equality $c^{\mathsf{T}}w = c^{\mathsf{T}}x$ forces $w_j = 0$ for every nonbasic index. The equality constraints then force $w_B = B^{-1}b = x_B$. Thus any optimal point equals x.

(c) Necessity at a nondegenerate optimal BFS

Suppose, for a contradiction, that $\bar{c}_j > 0$ for a nonbasic index j. Construct the basic direction by setting

$$d_j = 1, \qquad d_i = 0 \quad (i \in \mathcal{N} \setminus \{j\}), \qquad d_B = -B^{-1}A_j.$$

Then $Ad = 0$. Nondegeneracy means every component of x_B is strictly positive. Therefore for sufficiently small $t > 0$, all components of $x_B + td_B$ remain nonnegative; the nonbasic components are also nonnegative. Thus $x + td$ is feasible. But

$$c^{\mathsf{T}}(x + td) = c^{\mathsf{T}}x + t\bar{c}_j > c^{\mathsf{T}}x,$$

contradicting optimality. Hence every nonbasic reduced cost is nonpositive.

(d) A degenerate optimal point with a positive reduced cost

Consider the explicit LP

$$\max\ x_2 \qquad \text{s.t.} \quad x_1 + x_2 = 0, \quad x_1, x_2 \geq 0.$$

Here $A = (1\ \ 1)$ has rank 1. Choose x_1 as basic, so $B = (1)$ and $x_B = B^{-1}b = 0$. The associated full basic solution is $(0, 0)$, which is feasible and degenerate because its basic variable is zero.

The only feasible point is $(0, 0)$, so it is necessarily optimal, with value zero. Nevertheless

$$\bar{c}_2 = c_2 - c_B B^{-1} A_2 = 1 - 0 = 1 > 0.$$

The corresponding direction is $d = (-1, 1)$. No positive step is feasible because it makes x_1 negative. Thus a positive reduced cost does not contradict optimality at a degenerate BFS for an arbitrary representing basis.

(e) A zero reduced cost at a nondegenerate optimum

Choose a nonbasic index j with $\bar{c}_j = 0$ and construct the same basic direction as in part (c). Since x is nondegenerate, there is $\eta > 0$ such that $x + td$ is feasible for all $t \in [0, \eta]$. For each such t,

$$c^\mathsf{T}(x + td) = c^\mathsf{T}x + t\bar{c}_j = c^\mathsf{T}x.$$

These points are distinct because their jth coordinate equals t. Hence the feasible segment contains infinitely many optimal solutions.

Exercise 17. Simultaneous changes in a resource and a profit

For $\delta, \gamma \in \mathbb{R}$, consider

$$\begin{aligned} \max \quad & (4+\gamma)x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \le 12 + \delta, \\ & 3x_1 + x_2 \le 16, \\ & x_1, x_2 \ge 0. \end{aligned}$$

Introduce s_1, s_2 in constraint order and consider the ordered basis (x_1, x_2).

(a) Compute the full basic solution and the two nonbasic reduced costs as functions of (δ, γ).

(b) Determine the complete region of the (δ, γ)-plane where this basis is feasible and both reduced costs are nonpositive. Give the optimal-value formula on that region.

(c) Compute the optimal values for $(\delta, \gamma) = (0,0), (5,0), (0,2), (5,2)$. Does the change caused by applying both modifications equal the sum of the two changes obtained by applying each modification separately at $(0,0)$? Justify your answer numerically and algebraically.

(d) Set $(\delta, \gamma) = (20, -3)$. Does the specified basis satisfy the reduced-cost optimality test? Is its associated basic solution nevertheless optimal? Answer these two questions separately and prove both conclusions.

Solution to Exercise 17

(a) Basic solution and reduced costs

Order the full vector as (x_1, x_2, s_1, s_2). For the specified ordered basis,

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, \qquad B^{-1} = \begin{pmatrix} -1/5 & 2/5 \\ 3/5 & -1/5 \end{pmatrix}.$$

Setting $s_1 = s_2 = 0$ gives

$$\boxed{x = (4 - \tfrac{1}{5}\delta,\ 4 + \tfrac{3}{5}\delta,\ 0,\ 0).}$$

The basis multipliers are

$$y^{\mathsf{T}} = (4 + \gamma, 3)B^{-1} = (1 - \tfrac{1}{5}\gamma,\ 1 + \tfrac{2}{5}\gamma).$$

Since a slack column is a unit vector with zero objective coefficient, its reduced cost is the negative of the corresponding multiplier. Thus

$$\boxed{\bar{c}_{s_1} = -1 + \tfrac{1}{5}\gamma, \qquad \bar{c}_{s_2} = -1 - \tfrac{2}{5}\gamma.}$$

(b) Region where the basis satisfies both tests

The basic solution is nonnegative exactly for $-20/3 \le \delta \le 20$. The two reduced costs are nonpositive exactly for $-5/2 \le \gamma \le 5$. The required region is therefore the rectangle

$$\boxed{[-\tfrac{20}{3}, 20] \times [-\tfrac{5}{2}, 5].}$$

On it, this basis is optimal and

$$\boxed{z^\star(\delta, \gamma) = (4 + \gamma)(4 - \tfrac{1}{5}\delta) + 3(4 + \tfrac{3}{5}\delta) = 28 + 4\gamma + \delta - \tfrac{1}{5}\gamma\delta.}$$

(c) Separate versus simultaneous changes

All four parameter pairs lie in the rectangle. Their values are

(δ, γ)	Optimal value	Change from $(0,0)$
$(0,0)$	28	0
$(5,0)$	33	5
$(0,2)$	36	8
$(5,2)$	39	11

The simultaneous change is 11, whereas the sum of the separate changes is $5 + 8 = 13$. They differ because the value contains the interaction term $-\gamma\delta/5$, which equals -2 at $(5,2)$. The resource change alters the quantity x_1 to which the changed profit coefficient applies.

(d) Failure of the basis test does not settle point optimality

At $(\delta, \gamma) = (20, -3)$, the specified basis gives

$$x = (0, 16, 0, 0), \qquad \bar{c}_{s_1} = -\tfrac{8}{5}, \qquad \bar{c}_{s_2} = \tfrac{1}{5}.$$

Thus it *does not* satisfy the nonpositive-reduced-cost test. The basic solution is feasible but degenerate because the basic variable x_1 is zero.

The modified objective is $x_1 + 3x_2$. For any feasible point, the second inequality gives

$$x_1 + 3x_2 \leq 9x_1 + 3x_2 = 3(3x_1 + x_2) \leq 48.$$

The point $(0, 16)$ satisfies both resource constraints, $x_1 + 2x_2 = 32$ and $3x_1 + x_2 = 16$, and attains 48. It is therefore optimal despite the positive reduced cost for this basis. The nonnegative dual vector $(0, 3)$ gives the same upper-bound certificate.

In fact, equality in the displayed bound forces $x_1 = 0$ and $x_2 = 16$, so the optimizer is unique. Increasing the nonbasic variable s_2 from the specified BFS would change x_1 at rate $-2/5$; since $x_1 = 0$, no positive step in that direction is feasible.

Exercise 18. A new activity with an unknown profit

Consider first

$$\begin{aligned} \max \quad & 4x_1 + 5x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \le 16, \\ & x_1 + 3x_2 \le 18, \\ & x_1, x_2 \ge 0. \end{aligned}$$

A new activity $u \ge 0$ uses one unit of each resource and earns $p \ge 0$ per unit. The extended LP is

$$\begin{aligned} \max \quad & 4x_1 + 5x_2 + pu \\ \text{s.t.} \quad & 2x_1 + x_2 + u \le 16, \\ & x_1 + 3x_2 + u \le 18, \\ & x_1, x_2, u \ge 0. \end{aligned}$$

(a) Solve the original two-variable LP by simplex and give its optimal tableau.

(b) Incorporate u into that tableau. Determine the values of p for which the original basis remains optimal.

(c) For every $p \ge 0$, determine the optimal value and *all* optimal solutions of the extended LP. State every threshold value of p and treat the equality cases explicitly.

(d) Give a dual optimal solution in each parameter regime and verify that it certifies your answer in part (c).

Solution to Exercise 18

(a) Solving the original LP

Introduce slacks $s_1, s_2 \geq 0$. The initial tableau is

$$\begin{aligned} s_1 &= 16 - 2x_1 - x_2 \\ s_2 &= 18 - x_1 - 3x_2 \\ z &= 4x_1 + 5x_2 \end{aligned}$$

Choose x_2 to enter. Its bounds are 16 and 6, so s_2 leaves. This gives

$$\begin{aligned} s_1 &= 10 - \frac{5}{3}x_1 + \frac{1}{3}s_2 \\ x_2 &= 6 - \frac{1}{3}x_1 - \frac{1}{3}s_2 \\ z &= 30 + \frac{7}{3}x_1 - \frac{5}{3}s_2 \end{aligned}$$

Now x_1 has positive reduced cost. Its bounds are 6 from the s_1 row and 18 from the x_2 row, so s_1 leaves. The optimal tableau is

$$\begin{aligned} x_1 &= 6 - \frac{3}{5}s_1 + \frac{1}{5}s_2 \\ x_2 &= 4 + \frac{1}{5}s_1 - \frac{2}{5}s_2 \\ z &= 44 - \frac{7}{5}s_1 - \frac{6}{5}s_2 \end{aligned}$$

Thus $(x_1, x_2, s_1, s_2) = (6, 4, 0, 0)$ and the optimal value is 44.

(b) Incorporating the new activity

For the basis (x_1, x_2),

$$B^{-1} = \tfrac{1}{5}\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}, \qquad B^{-1}(1,1)^{\mathsf{T}} = (\tfrac{2}{5}, \tfrac{1}{5})^{\mathsf{T}}.$$

Hence the augmented tableau is

$$\begin{aligned} x_1 &= 6 - \tfrac{2}{5}u - \tfrac{3}{5}s_1 + \tfrac{1}{5}s_2, \\ x_2 &= 4 - \tfrac{1}{5}u + \tfrac{1}{5}s_1 - \tfrac{2}{5}s_2, \\ z &= 44 + (p - \tfrac{13}{5})u - \tfrac{7}{5}s_1 - \tfrac{6}{5}s_2. \end{aligned}$$

The old basis remains feasible. Its new reduced cost is nonpositive exactly when $p \leq 13/5$. Under the given restriction $p \geq 0$, the optimality range is $\boxed{0 \leq p \leq 13/5}$.

(c) All parameter regimes

For $0 \le p < 13/5$, all nonbasic reduced costs in the augmented tableau are strictly negative. Thus the unique optimum is $(x_1, x_2, u) = (6, 4, 0)$, with value 44.

At $p = 13/5$, the objective is 44 exactly when $s_1 = s_2 = 0$. Writing $t = u$, nonnegativity in the two rows gives $0 \le t \le 15$. All optima are therefore

$$(x_1, x_2, u) = (6 - \tfrac{2}{5}t,\ 4 - \tfrac{1}{5}t,\ t), \qquad 0 \le t \le 15.$$

For $p > 13/5$, u enters. The bounds are $u \le 15$ and $u \le 20$, so x_1 leaves. The resulting tableau is

$$\begin{aligned} u &= 15 - \tfrac{5}{2}x_1 - \tfrac{3}{2}s_1 + \tfrac{1}{2}s_2, \\ x_2 &= 1 + \tfrac{1}{2}x_1 + \tfrac{1}{2}s_1 - \tfrac{1}{2}s_2, \\ z &= 5 + 15p + \tfrac{13-5p}{2}x_1 + \tfrac{5-3p}{2}s_1 + \tfrac{p-5}{2}s_2. \end{aligned}$$

For $13/5 < p < 5$, every nonbasic reduced cost is strictly negative. The unique optimum is $(0, 1, 15)$, with value $5 + 15p$.

At $p = 5$, the objective row forces $x_1 = s_1 = 0$ at every optimum, while s_2 may vary from 0 to 2. Equivalently, all optima are

$$(x_1, x_2, u) = (0, t, 16 - t), \qquad 0 \le t \le 1,$$

and their common value is 80.

For $p > 5$, s_2 enters and x_2 leaves. The new tableau is

$$\begin{aligned} u &= 16 - 2x_1 - x_2 - s_1, \\ s_2 &= 2 + x_1 - 2x_2 + s_1, \\ z &= 16p + (4 - 2p)x_1 + (5 - p)x_2 - ps_1. \end{aligned}$$

Its nonbasic reduced costs are strictly negative. The unique optimum is $(0, 0, 16)$, with value $16p$. These objective rows also prove that the lists above contain *all* optimal solutions, not merely the BFSs.

(c) Optimal-value function

The complete value function can be written compactly as

$$\boxed{z^\star(p) = \begin{cases} 44, & 0 \le p \le 13/5, \\ 5 + 15p, & 13/5 \le p \le 5, \\ 16p, & p \ge 5. \end{cases}}$$

The formulas agree at both thresholds. At $13/5$ and 5, the full segments stated on the preceding page are optimal.

(d) Dual certificates in every regime

The extended dual is

$$\begin{aligned} \min \quad & 16y_1 + 18y_2 \\ \text{s.t.} \quad & 2y_1 + y_2 \ge 4, \\ & y_1 + 3y_2 \ge 5, \\ & y_1 + y_2 \ge p, \\ & y_1, y_2 \ge 0. \end{aligned}$$

For $0 \le p \le 13/5$, choose $y = (7/5, 6/5)$. The first two dual constraints are tight and $y_1 + y_2 = 13/5 \ge p$. The objective is $16(7/5) + 18(6/5) = 44$.

For $13/5 \le p \le 5$, choose

$$y = \left(\tfrac{3p-5}{2}, \tfrac{5-p}{2}\right).$$

Both coordinates are nonnegative. Moreover

$$2y_1 + y_2 = \tfrac{5p-5}{2} \ge 4, \qquad y_1 + 3y_2 = 5, \qquad y_1 + y_2 = p.$$

Its objective equals $16(3p-5)/2 + 18(5-p)/2 = 5 + 15p$.

For $p \ge 5$, choose $y = (p, 0)$. The dual left-hand sides are $2p \ge 4$, $p \ge 5$ and p, and its objective is $16p$.

Each dual feasible value matches the corresponding feasible primal value found in part (c). Weak duality therefore certifies every regime, including both thresholds.